



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2015

BIOLOGY

9648/01

HIGHER 2

31 August 2015

Paper 1 Multiple Choice

1 hour 15 mins

Additional Material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, pencil clips, highlighters, glue or correction fluid.

Write your name, centre number and index number on the Answer Sheet provided.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

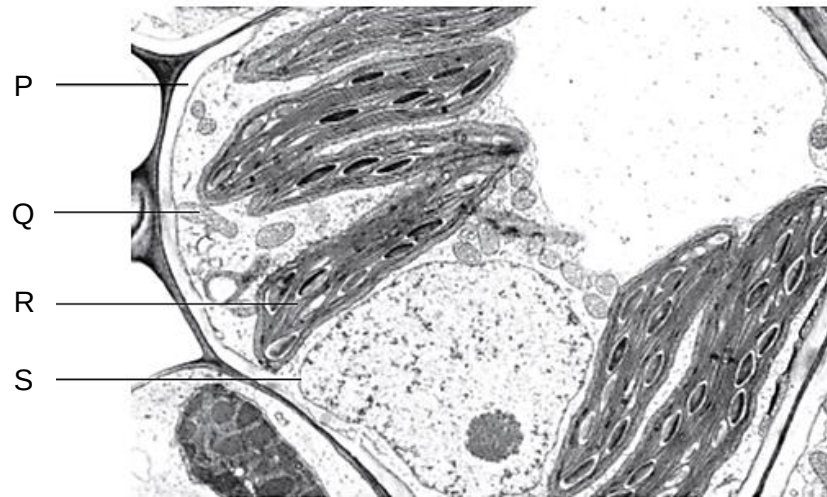
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Calculators may be used.

This question paper consists of **28** printed pages.

- 1 The electron micrograph of a cell is shown below.



Which of the following statements are true?

- 1 Structure P is found in all eukaryotic cells.
 - 2 Organelle Q contains hydrolytic enzymes.
 - 3 Organelle R contains starch.
 - 4 Organelle S contains heterochromatin but not euchromatin.
 - 5 Organelles Q, R and S contain RNA polymerase.
- A** 1 and 3 only
B 3 and 5 only
C 1, 3 and 4 only
D 2, 3 and 5 only
- 2 Three artificial cells of equal sizes, each surrounded by a partially permeable membrane, contained equal concentrations of different food reserves. One cell contained only fats, one cell contained only glycogen and one cell contained only glucose.

Which of the following shows the rate of expansion of these cells when immersed in distilled water, and their energy content?

(The sign $\approx$ means approximately equal and the sign $>$ means greater than.)

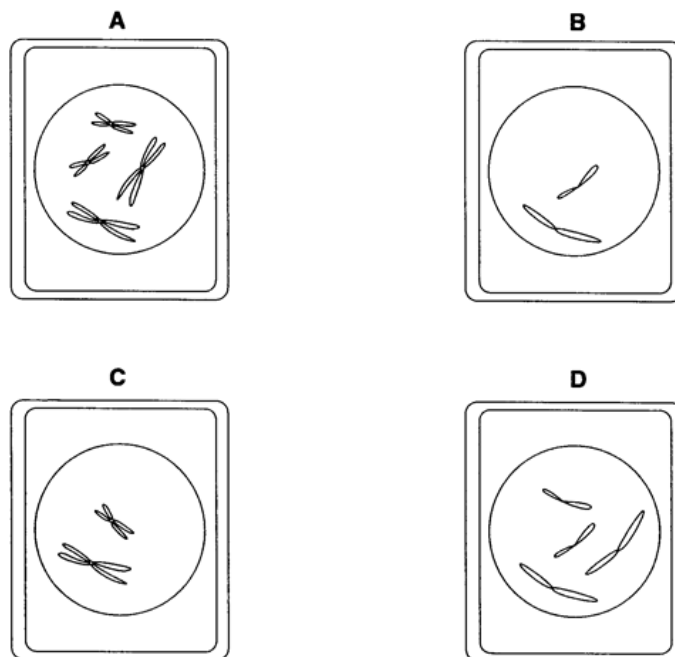
	Rate of expansion of cells	Respiratory energy content
A	glucose $\approx$ glycogen $>$ fat	fat $>$ glycogen $\approx$ glucose
B	glucose $\approx$ glycogen $>$ fat	fat $\approx$ glycogen $>$ glucose
C	glucose $>$ glycogen $>$ fat	fat $>$ glycogen $>$ glucose
D	glucose $>$ glycogen $>$ fat	fat $\approx$ glycogen $>$ glucose

- 3 Experiments were conducted to study the effects of cyanide on the rate of absorption of three monosaccharides by mammalian intestinal cells. Cyanide inhibits cytochrome c oxidases. The results of the experiments are shown in the table below.

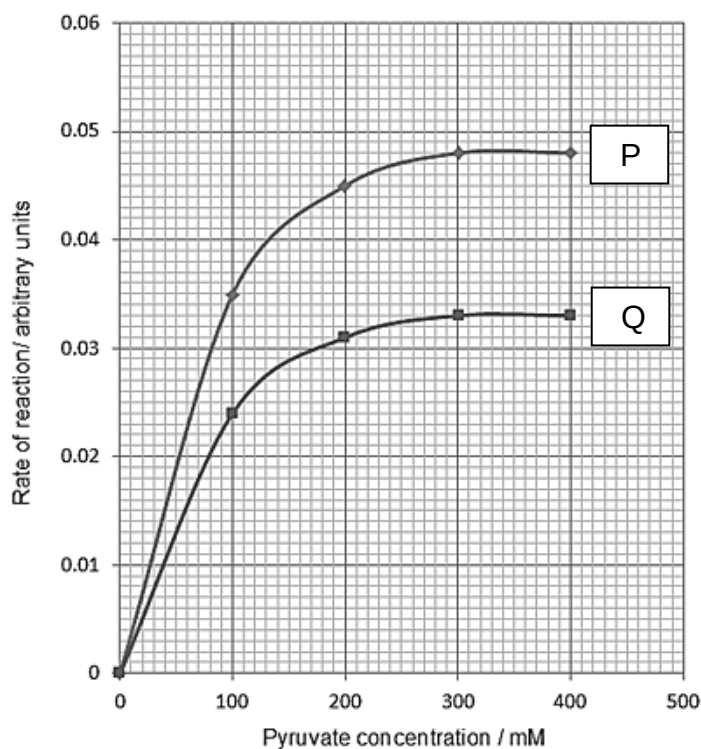
Monosaccharide	Relative rate of absorption by intestinal cells (arbitrary units)	
	In physiological saline	In physiological saline with added cyanide
Glucose	1.02	0.33
Xylose	0.33	0.33
Arabinose	0.27	0.28

Which of the following statements are consistent with these results?

- 1 Active transport is not involved in the uptake of xylose and arabinose.
 - 2 Xylose and arabinose enter the cells by facilitated diffusion.
 - 3 Aerobic respiration is necessary for higher rate of uptake of glucose.
 - 4 Extrinsic proteins are involved in the uptake of all the three monosaccharides.
- A** 1 and 2 only
B 2 and 4 only
C 1, 2 and 3 only
D 1, 3 and 4 only
- 4 A diploid cell contains four chromosomes. Which diagram shows the nucleus at prophase of mitosis after a previous meiotic division?



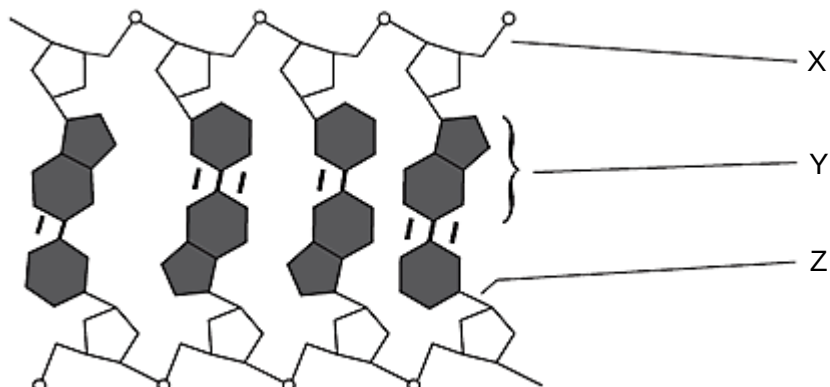
- 5 During the mitotic cell cycle, which of the following would result if cytokinesis does not occur after mitosis is completed?
- A Cells with two nuclei
 - B Cells with insufficient organelles
 - C Cells with twice the amount of genetic material but without nuclei
 - D Cells which are unusually small in size
- 6 Curve P shows the rate of a reaction catalysed by lactate dehydrogenase under optimum conditions. A change was made to the reaction and curve Q shows the effect of this change on the reaction rate.



Which factor, operating to a constant extent throughout the experiment, could result in curve Q?

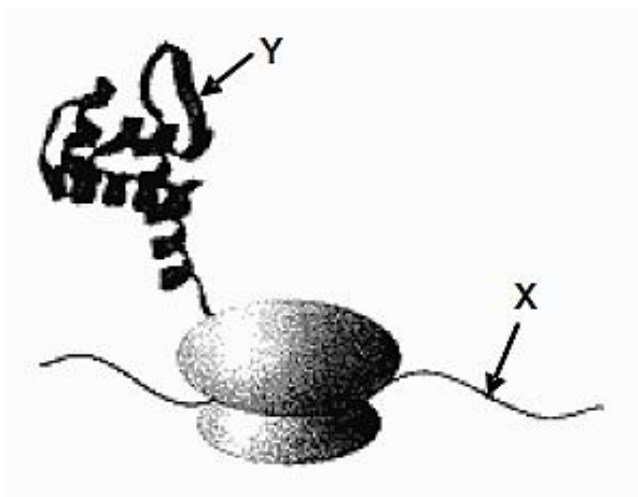
- A Addition of a compound that competes for the same binding site as pyruvate
- B Addition of an inhibitor that differs in 3D configuration from pyruvate
- C Addition of a co-enzyme such as NAD^+
- D An increase in enzyme concentration

- 7 The diagram below shows a short section of a DNA molecule. What are the correct labels for X, Y and Z?



	X	Y	Z
A	3' end	pyrimidine	hydrogen bond
B	3' end	purine	hydrogen bond
C	5' end	pyrimidine	covalent bond
D	5' end	purine	covalent bond

- 8 The diagram below shows a particular stage of protein synthesis.



Which of the following statements is true of molecules X and Y?

- A** The coiling of molecule Y is a direct result of the information on molecule X.
- B** Molecule X is double-stranded whilst molecule Y is single-stranded.
- C** In both molecules X and Y, the bonds between adjacent monomers of each molecule are phosphodiester bonds.
- D** The monomers of molecule X interact with the monomers of molecule Y through temporary hydrogen bonds.

- 9 The following table shows the mRNA codons for six different amino acids.

mRNA codons	amino acid
AAA AAG	lysine
AGA AGG CGG	arginine
GGU GGA GGC GGG	glycine
CCU CCA CCC CCG	proline
UGG	tryptophan
UAU UAC	tyrosine

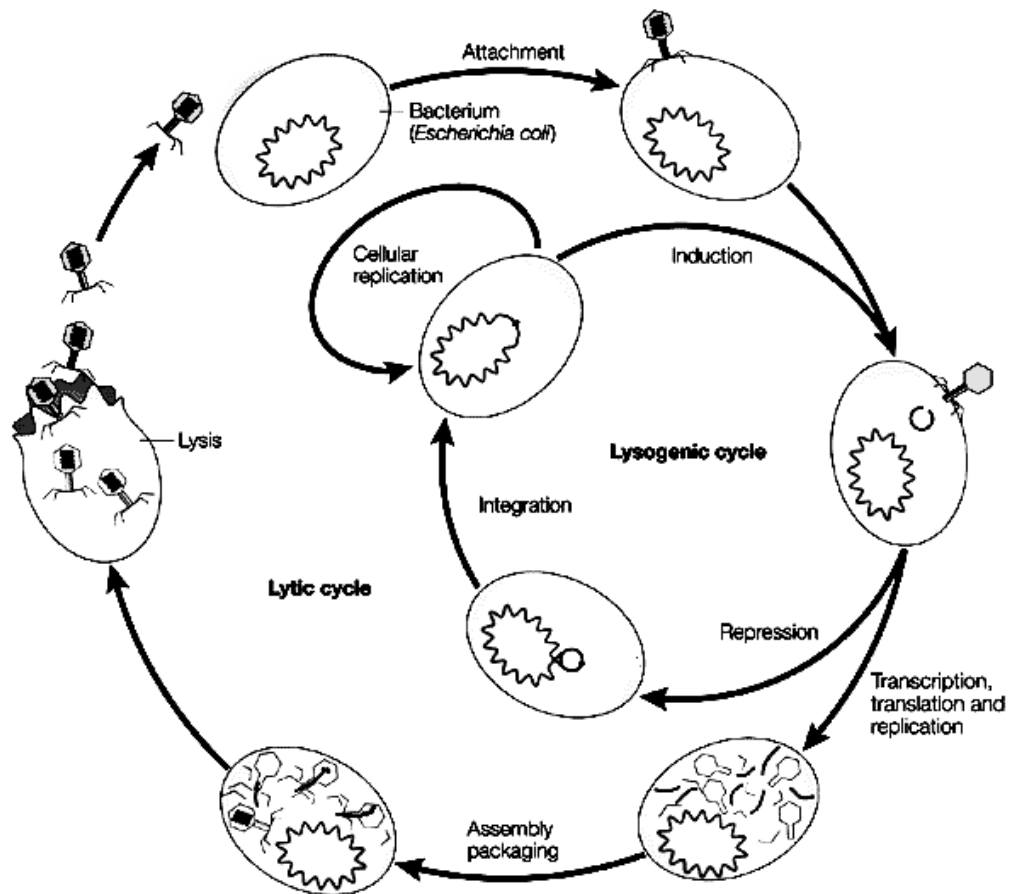
The base sequence of mRNA coding for part of a polypeptide is shown below.

U A U A A G A G G C C U U G G
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
 ↑
start reading

From the information provided, which of the predictions stated below is **not** true?

- A** The insertion of a nucleotide between positions 3 and 4 is expected to result in a greater change in the amino acid sequence than an insertion between positions 12 and 13.
- B** The deletion of a nucleotide at position 5 would result only in an alteration of the second amino acid in the chain.
- C** The substitution of a different nucleotide at position 12 would produce no alteration in the amino acid chain.
- D** The substitution of a different nucleotide at position 13 would result in the alteration of one amino acid.

10 The diagram below shows the reproductive cycles of a phage infecting a host bacterium.



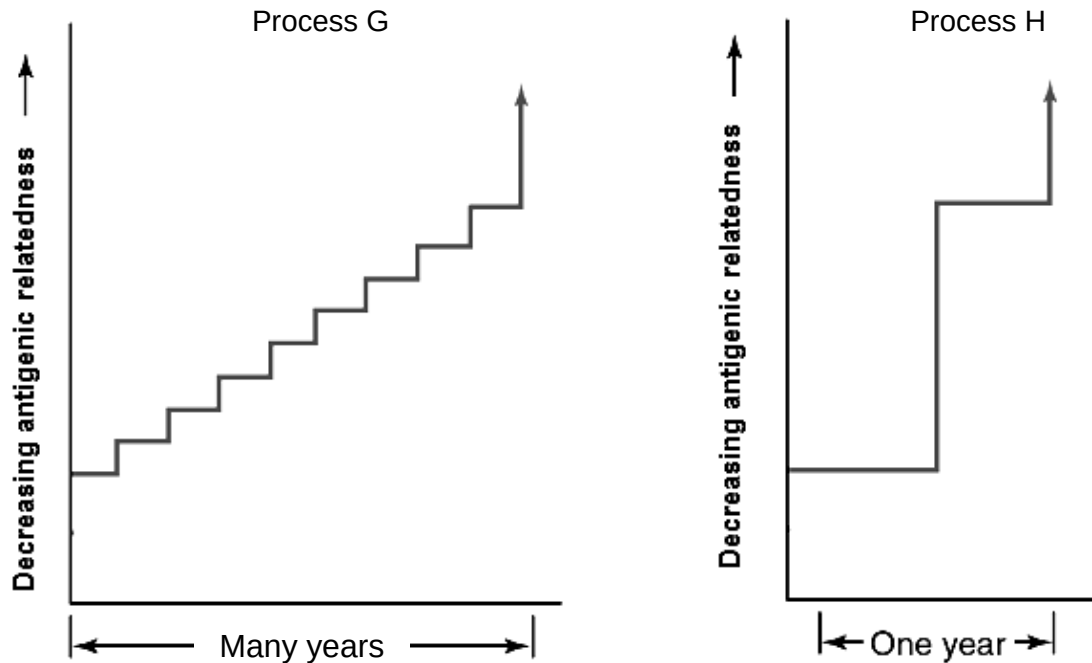
Some statements about the phages are listed as follows:

- 1 All types of phages are capable of undergoing lytic and lysogenic cycles.
- 2 A phage usually undergoes lysogenic cycle because a larger number of progeny phages can be produced rapidly as compared to the lytic cycle.
- 3 The release of lysozyme upon rupturing of lysosome leads to the osmotic lysis of host bacterium.
- 4 The phage gene codes for a repressor protein that prevents the expression of prophage.

Which of these statements about the reproductive cycles of a phage are **not** true?

- A 1 and 2 only
- B 3 and 4 only
- C 1, 2 and 3 only
- D 2, 3 and 4 only

- 11** Influenza infections are difficult to eliminate because of the antigenic changes that occur in haemagglutinin (HA), a surface glycoprotein of influenza virus. These changes are responsible for the formation of influenza viral strains with different levels of virulence. The figures below depict two processes (G and H) that contribute to the decreasing antigenic relatedness of HA between different viral strains over time.

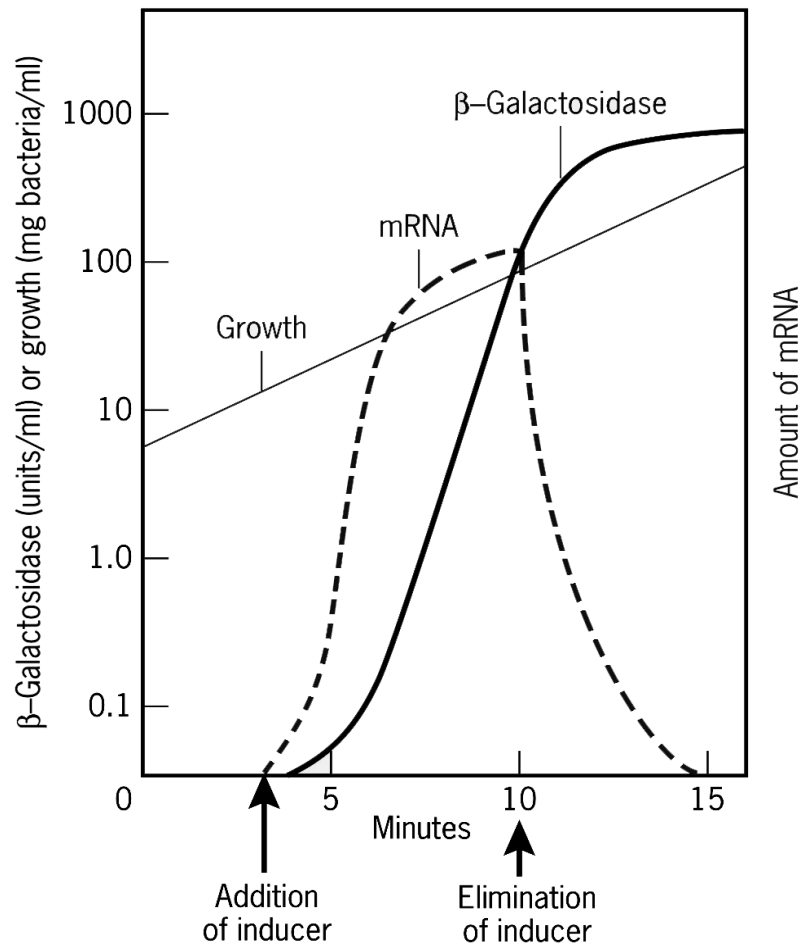


Which conclusions about processes G and H are valid?

- 1 G occurs due to a single point mutation in the gene coding for HA protein.
- 2 H may occur due to the genetic re-assortment of RNA genomes of human and avian influenza viruses.
- 3 G and H could only occur in an individual host who is simultaneously infected with at least two different strains of influenza virus.
- 4 The immune system of a host functions as a selective force for new antigenic variants of HA formed via process G.

- A** 1 and 2 only
B 1 and 3 only
C 2 and 3 only
D 2 and 4 only

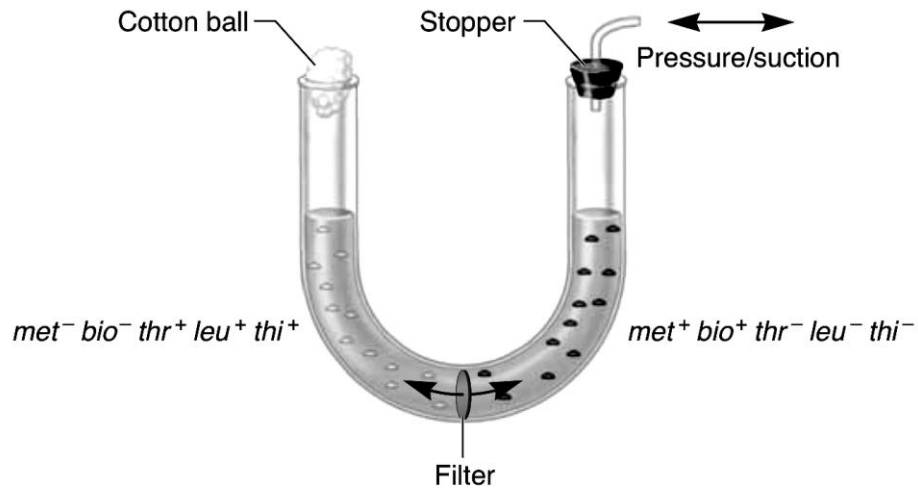
- 12 Prokaryotes such as *Escherichia coli* can be grown using culture medium containing essential elements and nutrients. The regulation of gene expression in prokaryotes can be examined by studying the kinetics of the inducible *lac* operon. The figure below shows the changes in the levels of β -galactosidase mRNA and enzyme, as well as the growth of bacteria when lactose (inducer) was added to and then removed from the bacterial culture.



Which of the following conclusions can be inferred from the figure?

- A The brief delay in the increase of β -galactosidase mRNA and enzyme levels upon addition of lactose is due to the need for conversion of lactose to allolactose, which will then bind to the *lac* repressor.
- B A rapid decrease in the level of β -galactosidase mRNA upon removal of lactose suggests that RNA polymerase is still able to bind to the promoter and transcribe the gene coding for β -galactosidase at a much lower rate.
- C The β -galactosidase mRNA is polycistronic as it contains coding sequences for structural genes such as lactose permease and lactose transacetylase.
- D The amount of β -galactosidase enzyme levels off upon removal of lactose, suggesting that the enzyme is stable and not degraded immediately even though it is no longer needed.

- 13** In 1950, Bernard Davis conducted an experiment involving two strains of bacteria with different nutritional requirements. A U-tube containing excess amounts of liquid medium with essential nutrients for growth of bacteria is shown in the figure below. At the bottom of the U-tube is a filter with pores small enough to allow the passage of genetic material only.



On one side of the filter, a mutant bacterial strain with a certain combination of nutritional requirements was added (the *met⁻ bio⁻ thr⁺ leu⁺ thi⁺* strain). On the other side, another mutant bacterial strain (the *met⁺ bio⁺ thr⁻ leu⁻ thi⁻* strain) was added. The application of alternating pressure and suction promoted the movement of liquid medium through the filter.

After incubation for 48 hours, bacteria from either side of the tube were placed on media lacking methionine, biotin, threonine, leucine and thiamine in order to select for the growth of wild-type bacterial strain. No bacterial colonies grew on these plates.

Based on the information given, which of the following statements best concludes the results of the experiment?

- A** Genetic transfer between different bacterial strains did not occur because they are unable to make physical contact with each other.
- B** Transformation did not take place as the bacteria on both sides of the filter stopped dividing due to over-crowding and lack of ATP production.
- C** Genetic transfer between different bacterial strains did not occur because bacteriophages were absent in the medium.
- D** The bacterial strains have different nutritional requirements as a result of loss-of-function mutations in genes coding for the synthesis of amino acids.

14 The following are descriptions of chromosomes.

- 1 Most species of organisms contain a circular chromosome, and it may be present in multiple copies.
- 2 A typical chromosome is a few million base pairs in length with several thousand genes which are closely packed together.
- 3 The ends of the chromosome can be lengthened using a RNA template.
- 4 A typical chromosome is tens of millions to hundreds of millions of base pairs in length and contains between a few hundred and several thousand different split genes.
- 5 Repetitive DNA sequences are found in greater abundance as compared to the coding DNA sequences.

Which of the features above are relevant to a eukaryotic chromosome?

- A** 1 and 4 only
- B** 2 and 3 only
- C** 1, 2 and 5 only
- D** 3, 4 and 5 only

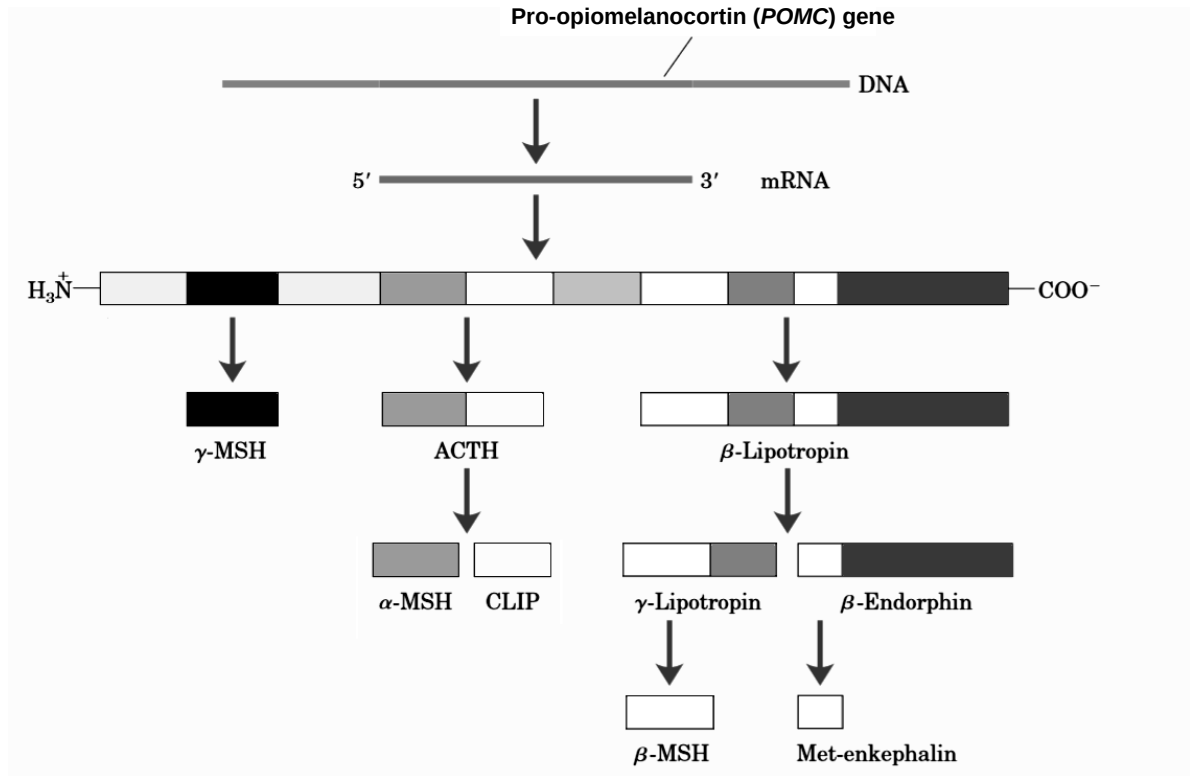
15 It is well established that steroid hormones enter the target cell directly by diffusing through the plasma membrane. An example is the glucocorticoid hormone which binds to intracellular glucocorticoid receptors. Prior to hormone binding, the glucocorticoid receptor is complexed with proteins known as Heat Shock Proteins (HSP).

Upon binding, a Nuclear Localisation Signal (NLS) on a glucocorticoid receptor is exposed, leading to the dissociation of HSP from the receptor. This also induces two glucocorticoid receptors to form a dimer which translocates into the nucleus. The dimer binds to a specific DNA sequence called Glucocorticoid Response Element (GRE) that controls the expression of a target gene. Since GREs are located near dozens of different genes, the transcription of these genes can be activated concurrently.

With reference to the information above, which of the following conclusions is valid?

- A** Steroid hormones are protein molecules which activate glucocorticoid receptors by binding to their allosteric sites.
- B** Glucocorticoid receptor complex is an activator protein, which binds to and activates Glucocorticoid Response Element that acts as an enhancer.
- C** Glucocorticoid Response Element is greatly amplified such that it recruits other regulatory transcription factors in stabilising the transcription initiation complex formed at the promoter of a target gene.
- D** Heat Shock Proteins are repressor proteins that mask the nuclear localization signal on glucocorticoid receptors and prevent it from binding to the silencer of a target gene.

- 16 The pro-opiomelanocortin (POMC) gene is expressed in the pituitary gland, the hypothalamus, the skin and the reproductive organs. This gene codes for a 285-amino acid polypeptide that undergoes processing to form nine different peptide hormones as shown in the schematic diagram below.



The processing of POMC polypeptide involves extensive proteolytic cleavage at sites shown to contain regions of basic amino acid sequences. The proteases that recognise these cleavage sites are tissue-specific. Multiple hormones are thus produced such as adrenocorticotrophic hormone (ACTH) and β -lipotropin in the anterior pituitary under the stimulation of corticotropin releasing hormone, as well as γ -lipotropin and β -endorphin in the intermediate lobe of pituitary gland under the stimulation of dopamine. The terminal residues of these peptide hormones are often glycosylated or acetylated.

Which of the following statements is true?

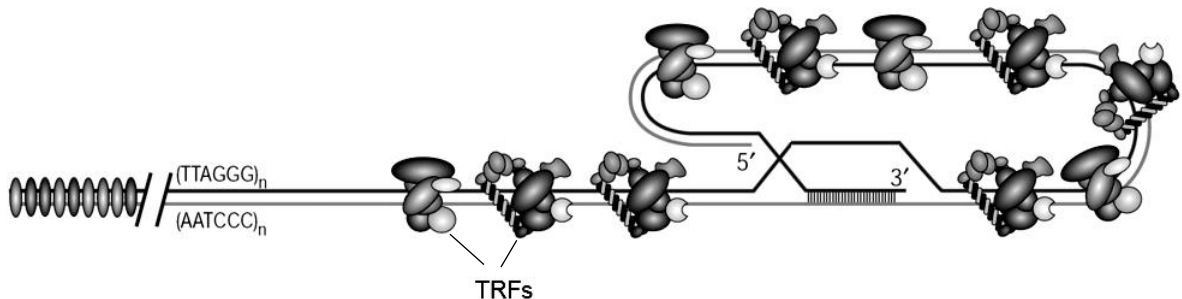
- A** The types of hormones formed in a specific tissue depend on the control of gene expression at the translational level.
- B** Specific chemical signals are required for the formation of unique peptide hormones in different tissues.
- C** The different hormones are formed by alternative splicing.
- D** Parts of the amino acid sequence of POMC polypeptide can undergo rearrangement to form peptide hormones of varying length.

- 17** In a mouse model experiment, deletion of the multi-drug resistance (*mdr*) gene in liver cells leads to the development of liver cancer. Loss of *mdr* gene and its encoded protein leads to the accumulation of bile acids that initiates liver inflammation, a process that recruits white blood cells to the target site.

These white blood cells secrete Tumour Necrosis Factor- α (TNF- α) that binds to the corresponding receptor on liver cells, which in turn activates a specific transcription factor. Activation of this transcription factor has been shown to result in the elevated levels of an anti-apoptotic protein and a growth-promoting protein produced by liver cells. These proteins are involved in the progression of liver cells to turn cancerous.

Based on the information above, which of the following can be concluded?

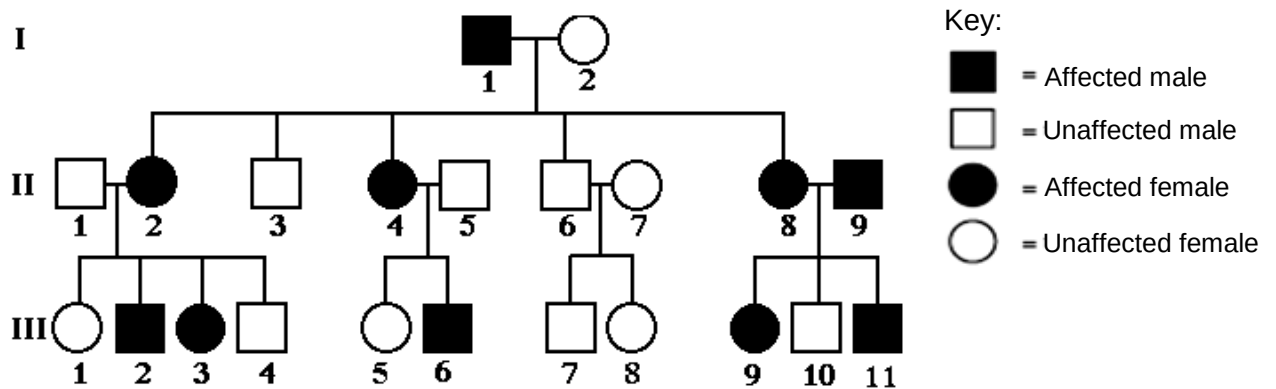
- A** *mdr* gene is a tumour suppressor gene which codes for a protein involved in apoptosis.
 - B** The progression to liver cancer requires mutations in proto-oncogenes and tumour suppressor genes.
 - C** The progression to liver cancer can occur when a loss-of-function mutation in the *mdr* gene results in the over-expression of the other proto-oncogenes.
 - D** The metastasis of liver cancer is accelerated with the recruitment of more white blood cells due to elevated levels of TNF- α .
- 18** A telomere is comprised of a 230-kb array of duplex TTAGGG repeats. This DNA sequence exists as a telomere loop, in which the 3' overhang folds back on itself. Proteins known as TTAGGG repeat-binding factors (TRFs) are associated with the duplex repeats.



Which of the statements below explains the significance of the formation of the telomere loop?

- A** Telomeres protect the ends of chromosomes from enzymatic degradation or fusion with other chromosomes.
- B** Telomeres are highly conserved across all eukaryotic species with little variation in the base sequences.
- C** Telomeres are repetitive sequences that are permanently cross-linked by proteins such that the chromosome ends will not be truncated after every cell division.
- D** Telomeres are DNA-protein complexes that have regulatory functions in preserving the length of genes found at the chromosomal ends.

- 19 A diagram shows the pedigree of Rett's disease, a genetic neurological disorder which results in autism-like symptoms.



Which of the following statements can be concluded based on the diagram above?

- A This genetic disorder is a X-linked dominant trait.
 B This genetic disorder is due to an autosomal recessive allele.
 C Individual II-1 must be heterozygous for the mutant allele as III-2 and III-3 are affected with the genetic disorder.
 D Individual III-11 was affected with the genetic disorder as a result of genetic mutation during his lifetime.
- 20 The colour of roses is controlled by one gene. There are three possible phenotypes – red, white, and white roses with red patches. Two crosses were conducted.

Cross 1	Homozygous red roses and homozygous white roses were crossed.
Cross 2	White roses with red patches from the F ₁ generation were selfed.

Which of the following shows the correct set of ratios in the offspring?

	Cross 1				Cross 2		
	Red	Red and white	White		Red	Red and white	White
A	1	2	1		1	2	1
B	0	1	0		1	2	1
C	0	1	0		3	0	1
D	1	0	0		3	0	1

- 21** Corn snakes are brightly-coloured snakes which show a variety of colouration. A scientist crossed two doubly heterozygous corn snakes and predicted that there would be four phenotypic classes – normal, amelanistic, anerythristic and snow in the ratio 9:3:3:1. Out of the offspring produced, 65 were normal, 12 were amelanistic, 15 were anerythristic and 8 were snow snakes.

The formula for χ^2 and the table of probabilities are given as follows:

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \nu = c - 1$$

where Σ = 'sum of...'

O = observed 'value'

E = expected 'value'

ν = degrees of freedom

c = number of classes

Degree of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Which of the following statements is correct?

- A** The calculated chi-squared value is greater than the critical chi-squared value.
- B** The probability that the difference between observed and expected values is not due to chance is more than 90%.
- C** The probability that the difference between observed and expected values is due to chance is more than 10%.
- D** There is a low probability that the difference between the observed and expected values is due to chance.

- 22 In a plant, three genes are known to be linked. The table below gives the crossover values (COVs) obtained from crosses involving pairs of these genes.

Pair of linked genes	COV
Colour of flowers (P) x Appearance of fruit (Q)	43
Appearance of fruit (Q) x Presence of prickles (R)	17
Presence of prickles (R) x Colour of flowers (P)	26

- 1 The chance of crossing over occurring between genes P and R will be lower than that of Q and R.
- 2 Genes Q and R are 17 map units apart on the same chromosome.
- 3 The order of the genes on the chromosome is P – R – Q.
- 4 The COV values are obtained based on the proportion of recombinant offspring.

Based on the information, which of the above statement(s) is/are **false**?

- A 1 only
 B 1 and 2 only
 C 2 and 3 only
 D 2 and 4 only
- 23 The inheritance of flower colour in sweet pea plants is controlled by the interaction between two different genes. The biochemical pathway below illustrates the synthesis of purple pigments in these sweet pea plants.

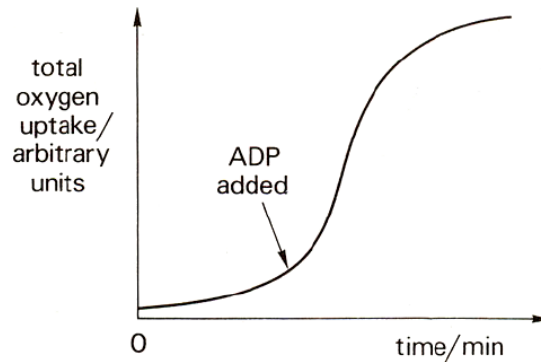


Which of the following statements are **false**?

- 1 Recessive alleles at either gene loci in the homozygous recessive condition produce the same phenotype without a cumulative effect.
- 2 Inheritance of flower colour in sweet pea is an example of dominant epistasis.
- 3 Self-pollination of a doubly heterozygous sweet pea plant results in three different phenotypes.
- 4 Self-pollination of a doubly heterozygous sweet pea plant results in an offspring phenotypic ratio of 15:1.

- A 1 and 3 only
 B 2 and 4 only
 C 2, 3 and 4 only
 D 1, 2 and 4 only

- 24 The graph shows the total oxygen uptake by a sample of homogenised liver tissue over a period of time.



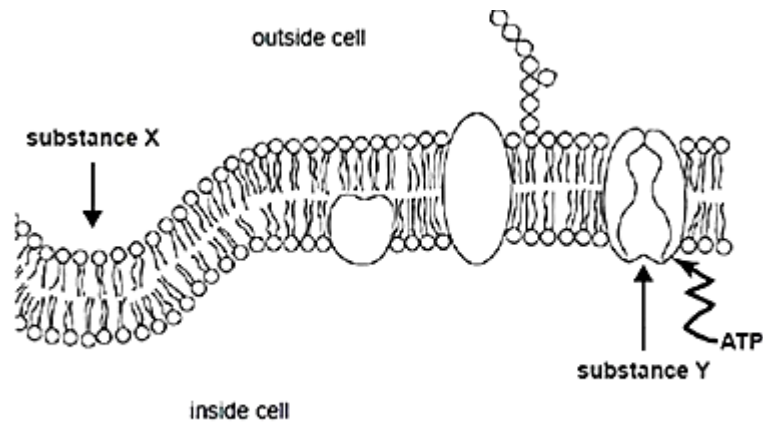
Which statement explains the sharp rise in oxygen uptake after the addition of ADP?

- A Electron transport is coupled to the phosphorylation of ADP.
 - B More protons move directly across the phospholipid bilayer of the cristae, generating the proton pool in the intermembrane space.
 - C Slower utilisation of reduced NAD and FAD causes molecular oxygen to be reduced to water.
 - D Oxidative phosphorylation occurs as protons are pumped back into the matrix through ATP synthase, phosphorylating ADP.
- 25 In cyclic photophosphorylation, when the reaction centre of Photosystem I absorbs light, its electron is excited and accepted by an electron acceptor X before it is passed on to a series of electron carriers.

Which of the following would **not** take place in the presence of an inhibitor of electron acceptor X?

- A Carbon dioxide gas diffuses through the stomata at a faster rate.
- B Less oxygen gas is released.
- C Less protons will be found in the thylakoid space.
- D Reduced NADP is produced in smaller amounts.

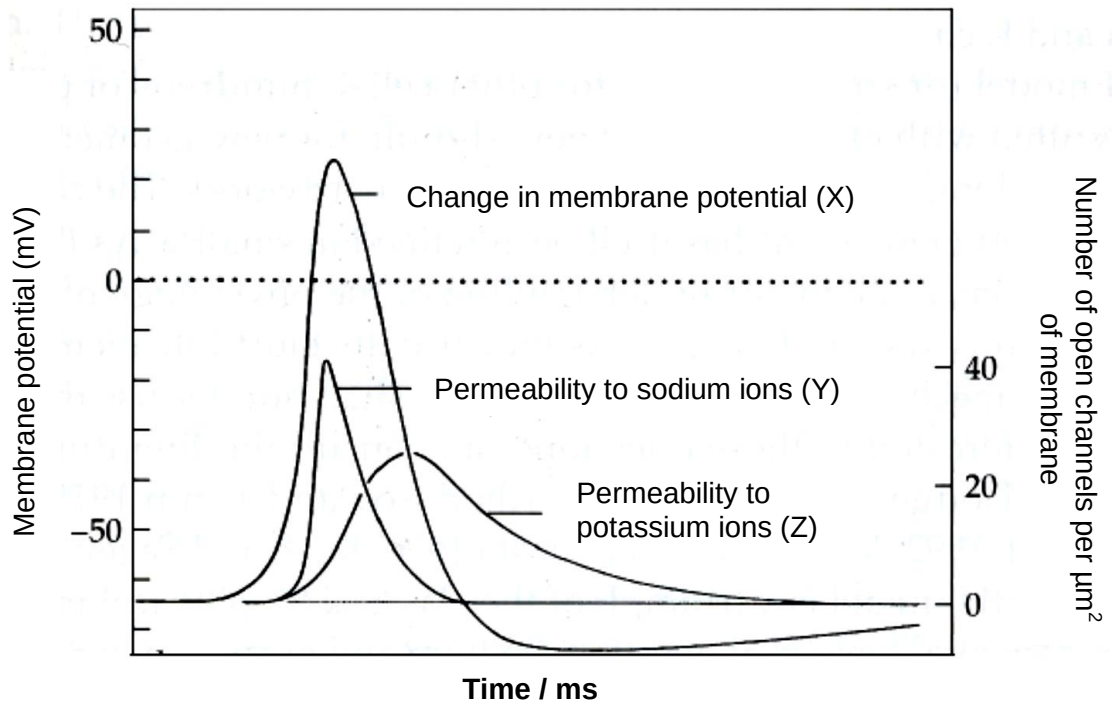
- 26 The figure below shows a cross section of the plasma membrane in a typical mammalian cell. The substances labelled X and Y are about to be transported across the membrane in the directions shown by the arrows ($\longrightarrow$).



Which of the following rows is correct?

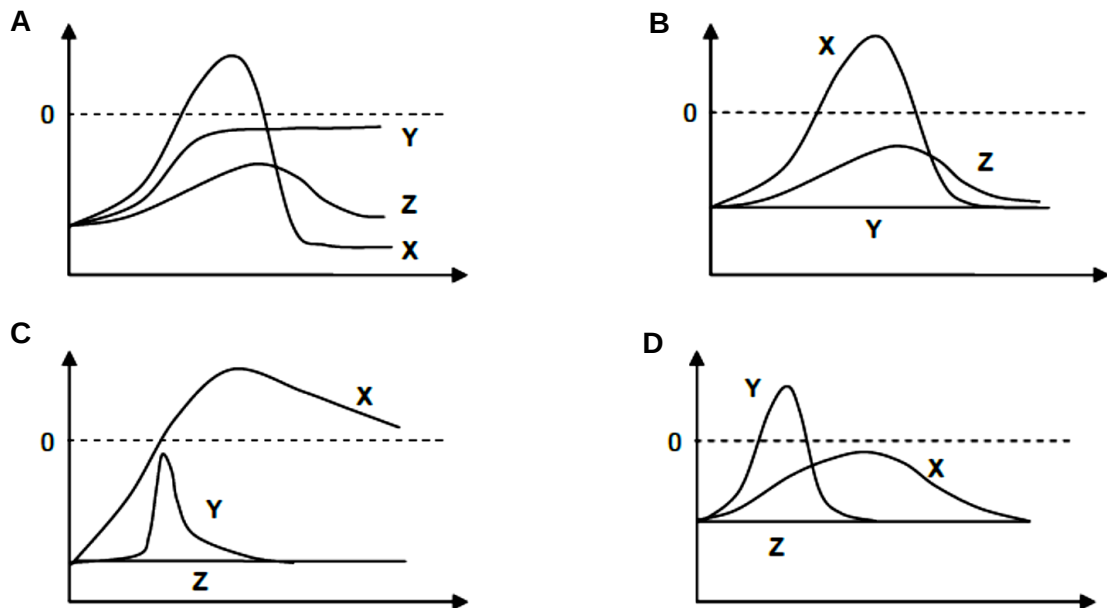
	Substance X			Substance Y		
	type of movement	nature of molecule	size of molecule	type of movement	nature of molecule	size of molecule
A	diffusion	lipid soluble	small	facilitated diffusion	lipid soluble	large
B	diffusion	water soluble	small	active transport	lipid soluble	small
C	endocytosis	lipid soluble	large	facilitated diffusion	charged	large
D	endocytosis	water soluble	large	active transport	charged	small

- 27 The diagram below highlights the changes in membrane potential and permeability to specific ions in a neurone during an action potential.

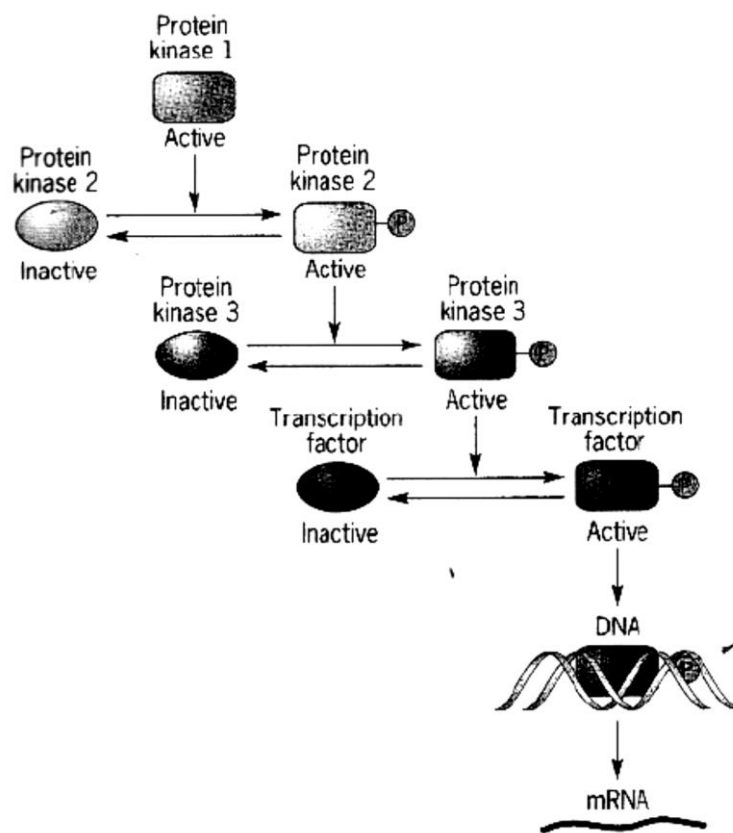


Maurotoxin is a neurotoxin released by scorpions that blocks the efflux of cations.

Which of the following graphs correctly represents the change upon addition of maurotoxin to a neurone during an action potential?



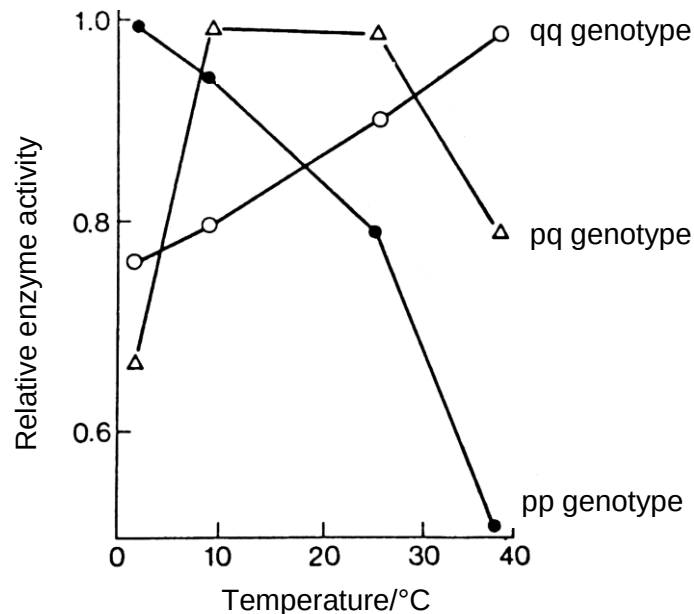
- 28 A diabetic patient has accidentally given himself an overdose of insulin. Which of the following treatments should be given to this patient?
- A A drug that blocks the action of glucagon.
 - B A drug that results in a slower breakdown of ingested carbohydrates.
 - C A drug that stimulates the pancreatic beta cells.
 - D Glucose solution to be administered by injection.
- 29 The figure below shows part of a signalling pathway in a cell. The pathway leads to the activation of a gene that codes for an inhibitor of a cyclin-dependent kinase responsible for moving a cell into the S phase of the cell cycle.



Which of the following statements correctly describes how the rate of cell division will be affected if a mutation results in a non-functional Protein kinase 3?

- A Cell division would proceed at a rapid rate.
- B Cell division would proceed very rapidly initially, followed by a sharp decrease in the rate as the cell cycle progresses.
- C Cell division would proceed, but at a decreased rate.
- D Cell division would be completely inhibited.

- 30** In the North American catfish *Catostomus clarki*, two alleles, represented by p and q, control the synthesis of a vital enzyme. The three possible genotypes (pp, pq, qq) lead to the synthesis of variations of the same enzyme with different optimal temperatures as shown in the graph below.



When the mean annual temperature is 5°C, which of the following statements is correct?

- A** Frequency of allele p in the gene pool will increase.
B Frequency of allele q in the gene pool will increase.
C Allele p will become dominant and allele q will become recessive.
D The heterozygotes will have an advantage over the homozygotes.
- 31** A study was done on the Dunkers population in Pennsylvania, America. Dunkers are a religious sect which emigrated from Germany to America in the 18th century. Each Dunker community was made up of about 100 families and many of these have married exclusively amongst their own members. Blood group analyses produced the following results.

Population	Proportion with blood group A / %
Indigenous Pennsylvanians	42
Indigenous West Germans	42
Dunkers	60

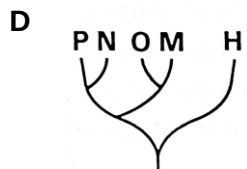
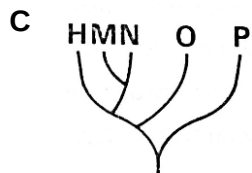
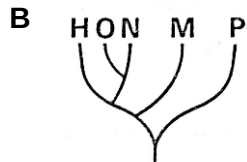
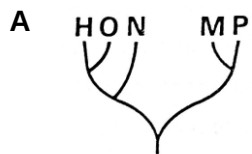
Which of the following best explains the above results?

- A** Natural selection
B Mutation
C Founder effect
D Non-random marriages

- 32 Human blood, when mixed with antibodies to human blood, will give maximum precipitation. When other animals' blood was mixed with antibodies to human blood, the following experimental results were obtained.

Species	Human	M	N	O	P
Percentage of precipitation / %	100	37	75	79	17

If the human species is denoted by the symbol H, which of the following phylogenetic trees would reflect the above results?

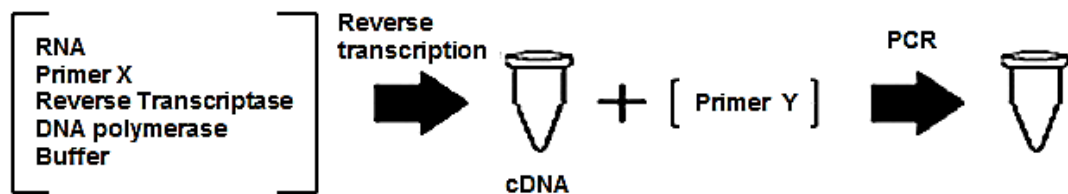


- 33** Nucleoside inhibitors are an example of drugs used in controlling the human immunodeficiency virus (HIV) which causes acquired immune deficiency syndrome (AIDS). These inhibitors compete with DNA nucleosides for the active site of the reverse transcriptase of HIV. This enzyme is vital to ensure the reproductive success of HIV. One of these nucleoside inhibitors, 3TC, is a molecular analogue of cytosine.

Soon after 3TC is first given to the patient, a modified form of reverse transcriptase which binds to cytosine but not 3TC, will appear at a very low frequency in the HIV population in a human body. The percentage of this drug-resistant HIV in the patient will then increase from the time the drug is first given to the patient, to 100% about three weeks later.

Which of the following statements best supports the above information?

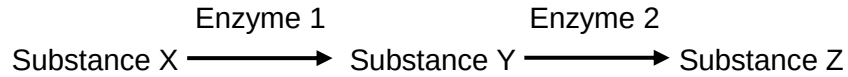
- A** The presence of 3TC triggers the modification of reverse transcriptase which only binds to cytosine, and not 3TC.
 - B** The presence of 3TC causes the virus to select for reverse transcriptase which only binds to cytosine, and not 3TC.
 - C** The virus increases the expression of the gene coding for the modified form of reverse transcriptase.
 - D** High mutation frequency in HIV would have resulted in the production of HIV variants, some with modified reverse transcriptase.
- 34** Reverse transcriptase polymerase chain reaction (RT-PCR) is a technique which couples the process of reverse transcription to polymerase chain reaction. After extracting mRNA from a target eukaryotic cell, the extracted sample is subjected to RT-PCR as follows:



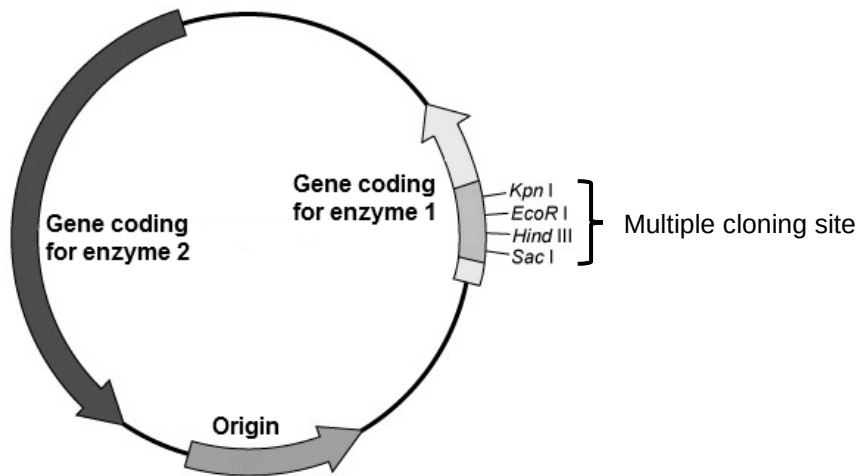
Which of the following statements describing RT-PCR is **false**?

- A** Primer X used for reverse transcription can be an oligonucleotide sequence of repeating thymines.
- B** Both primer X and primer Y used for reverse transcription and for PCR are DNA primers.
- C** RT-PCR can lead to the exponential increase in the quantity of mRNA for analysis.
- D** RT-PCR can be used to study the expression of oncogenes and tumour suppressor genes in cancer cells.

- 35** The metabolic pathway shown below is utilised by a species of bacteria to produce substance Z, which is essential for the replication of the bacterial chromosome. When provided with substance X or substance Y, wild-type bacteria are able to synthesise substance Z.



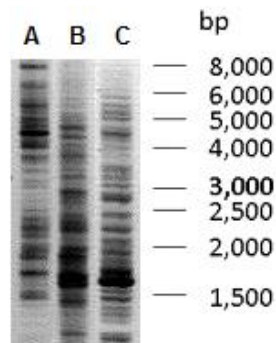
Mutant bacteria, lacking the genes coding for both enzyme 1 and 2, were genetically modified to contain the human insulin gene. The following plasmid was used as a vector for transformation.



The transformed cells were plated onto an agar plate containing only substance Y. Which of the following replica plates have to be prepared in order to identify the colony containing recombinant bacteria?

- A** Replica plate containing substance X only
- B** Replica plate containing substance X and Y
- C** Replica plate containing substance Y and Z
- D** Replica plate containing substance Z only

- 36** DNA fingerprinting can be used to establish evolutionary relationships between different species. Restriction fragment length polymorphism (RFLP) analysis is performed on the DNA samples from three species: A, B and C. Based on the results, species B and C are said to be more closely related to each other, than to species A.



Which of the following assumptions have to be made in order to form the above conclusion?

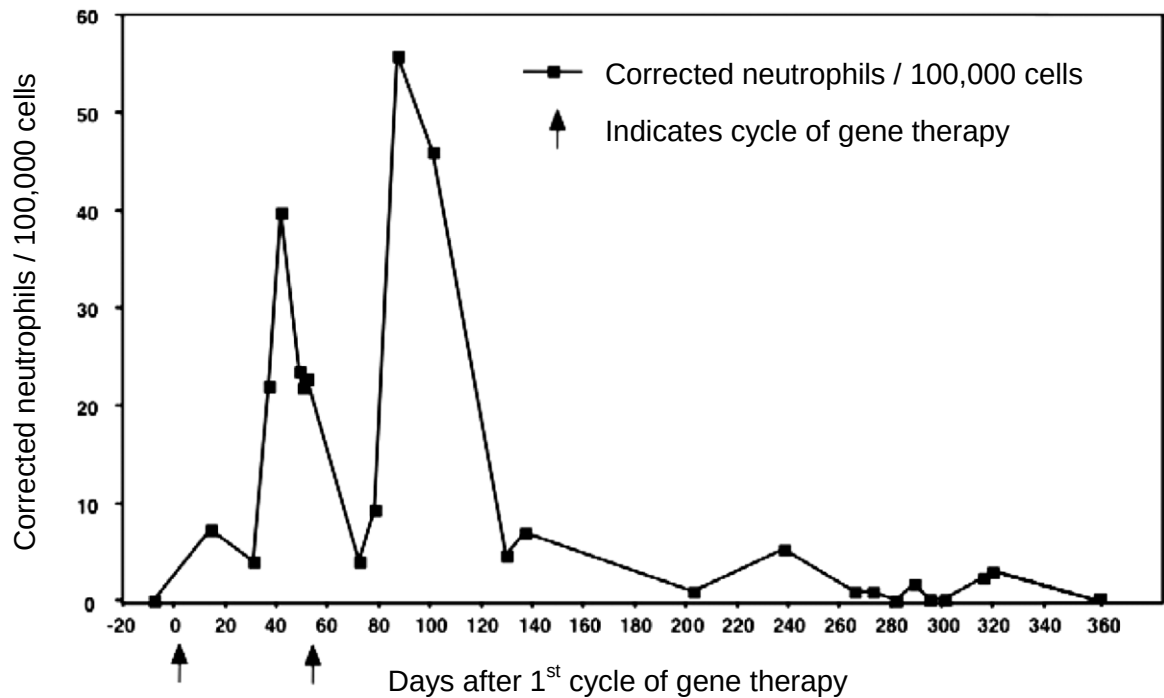
- 1 The three DNA samples are treated with the same set of restriction enzymes.
 - 2 The same DNA probe is used for the visualisation of DNA bands in all three lanes.
 - 3 Closely related species have lower sequence homology in their DNA.
 - 4 Closely related species are more likely to share the same number and position of restriction sites.
- A** 1 and 2 only.
- B** 1 and 3 only.
- C** 1, 2 and 4 only.
- D** 2, 3 and 4 only.
- 37** After the human genome was fully sequenced in 2003, scientists launched a second project to study the human genome, The Encyclopedia of DNA Elements (ENCODE). The goal of ENCODE is to build a comprehensive list of functional elements in the human genome, including regulatory elements that control gene expression, with the aim of better understanding human biology and to predict potential risk of diseases.

The project has identified previously undiscovered DNA regulatory elements, providing insights into the organization and regulation of human genes, and how differences in DNA sequence could influence the development of diseases.

Which of the following ethical concerns addressed by the Human Genome Project does **not** directly apply to the ENCODE project?

- A** Fairness in access to advanced genomic technologies
- B** Uncertainties associated with gene tests for susceptibilities and complex conditions
- C** Stigmatization and discrimination due to an individual's genetic makeup
- D** Health and environmental issues concerning genetically modified (GM) foods and microbes

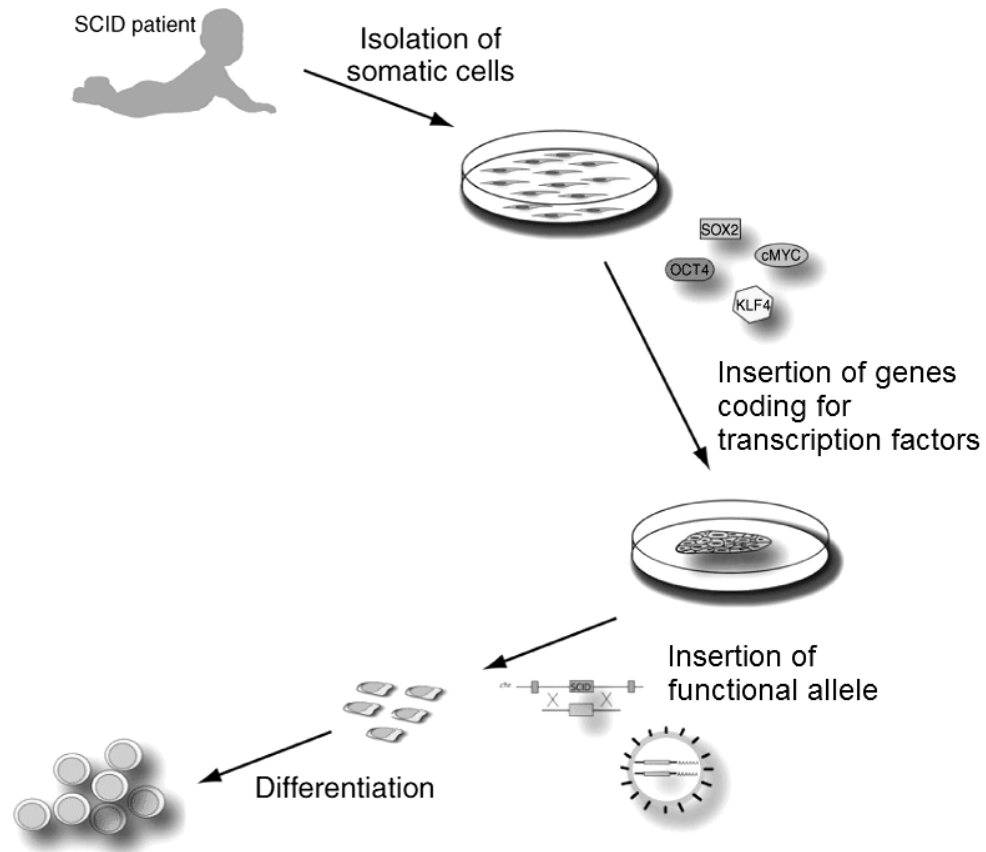
- 38** Chronic granulomatous disease (CGD) is a genetic disorder where white blood cells, such as neutrophils, are unable to perform phagocytosis of ingested pathogens. A patient with CGD received two cycles of gene therapy, where murine (mouse) virus vectors were used to deliver a copy of the normal allele. The number of functioning neutrophils (corrected neutrophils) was measured in the patient after gene therapy, as shown in the following diagram.



Which of the following statements may explain the results of the gene therapy as shown?

- A** The functional allele was introduced into haematopoietic stem cells instead of neutrophils.
- B** The introduced allele entered the host nucleus and remained as a separate molecule from the host chromosomes.
- C** The viral vector was unable to bind to host cells due to the lack of receptors which are complementary to viral glycoproteins.
- D** After the first cycle of gene therapy, the host has developed immunity against the viral vector.

- 39** In 2006, scientists have discovered a means to convert somatic cells, such as skin cells, into pluripotent stem cells. By inserting four specific genes which code for transcription factors, somatic cells are able to de-differentiate and achieve pluripotency. Such pluripotent stem cells may one day be used for gene therapy to treat diseases such as severe combined immunodeficiency (SCID), as shown in the following diagram.



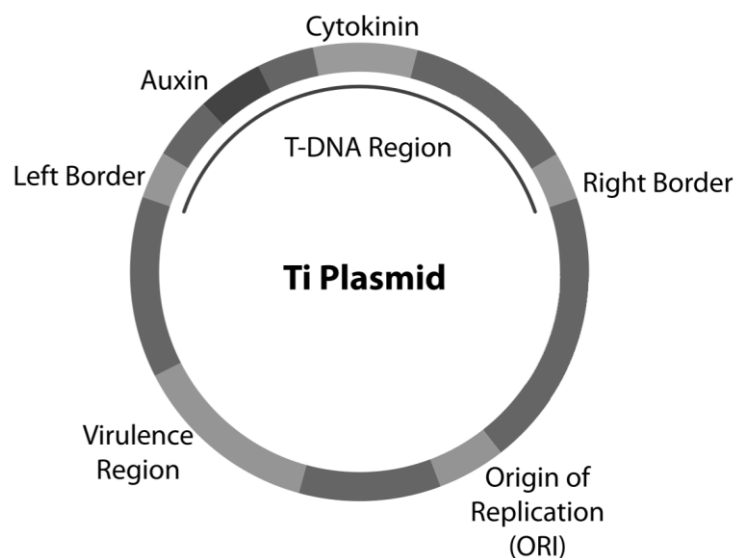
What is one benefit of using these novel pluripotent stem cells over embryonic stem cells?

- A** A functional copy of the allele could be inserted into these novel stem cells using a retroviral vector during gene therapy.
- B** These novel stem cells have the potential to differentiate into more types of cells, hence they can be used to treat a greater variety of diseases.
- C** These novel stem cells will not be rejected by the host after being transplanted back into the body.
- D** These novel stem cells are able to maintain the undifferentiated state for long periods of time due to the insertion of additional genes.

- 40** *Agrobacterium tumefaciens* is a soil bacterium that has been used in biotechnology for the insertion of foreign genes into the plant cells that it infects. It contains the Ti plasmid, which is shown in the diagram below.

After the infection of a plant cell, the T-DNA region is excised from the plasmid and introduced into the plant cell, leaving the remainder of the plasmid in the bacterial cell. The T-DNA subsequently enters the nucleus and is integrated into the host cell genome.

The T-DNA region contains genes coding for auxin and cytokinin. These plant growth regulators stimulate plant cell proliferation, leading to tumour formation in the infected plant.



Genes in the Ti plasmid which induce tumour formation are replaced with foreign genes which can confer pest resistance. Such modified plasmids are used to create transgenic sugarcane cultivars. Which of the following implications applies to the creation of these cultivars?

- A** It has the potential to benefit mankind due to the improved quality of food crops.
- B** When the transgenic sugarcane is consumed, genes responsible for the virulence of *Agrobacterium tumefaciens* may be passed on to other bacteria in the environment.
- C** The long-term consumption of transgenic sugarcane containing elevated levels of auxin and cytokinin may have unknown effects on human health.
- D** The large-scale use of transgenic sugarcane in agriculture can have undesirable effects on ecological balance.

End of Paper